O-E-950A
March 10, 1975
SUPERSEDING
O-E-00950
June 16, 1964

FEDERAL SPECIFICATION

EXTRACT, QUEBRACHO

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of cold-water-soluble quebracho extract (see 6.1).

1.2 Classification. Quebracho extract shall be of the following types, as specified (see 6.2):

Type I - Powdered.
Type II - Solid (lump).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC; Atlanta, Chicago, Kansas City, MO; Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

FSC 6810

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Leather Chemists Association (ALCA):

A20 - Total Solids and Water.
A21 - Soluble Solids and Insolubles.
A22 - Nontannins and Tannin.

(Application for copies should be addressed to the American Leather Chemists Association, Campus Station - Location 14, Cincinnati, OH 45221.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, NW., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. Type I quebracho extract shall be in powdered form, either ground or spray-dried. Type II quebracho extract shall be in solid (lump) form.

3.2 Chemical requirements. Quebracho extract shall conform to the chemical requirements of table I when tested as specified therein.

TABLE I. Chemical requirements for quebracho extract

Type	I	II	Test paragraph
Requirement			
Water, percent maximum	14	18	4.3.2
Insolubles, percent maximum	1	1	4.3.3
Tannin, percent minimum	70	66	4.3.4

3.3 Workmanship. Quebracho extract shall be a uniform color, ranging from pale yellow to pale brown, visibly free from foreign matter, and suitable for the intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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glass rod with water. Add several drops of bromothymol blue indicator (0.1 per-

4.2 Quality conformance inspection.

4.2.1 Examination of preparation for delivery.

4.2.1.1 Lot. For this examination, a lot shall be quebracho extract, fully packed, marked, and offered for examination at one time. Lot size shall be expressed in number of filled unit containers.

4.2.1.2 Sampling plan. Sample unit for this examination shall be one unit container. Sampling shall be conducted in accordance with MIL-STD-105. Sample size, acceptance and rejection numbers shall be determined by general inspection level II and acceptable quality level (AQL) 6.5 percent defective, respectively. Defects shall be scored in accordance with table II.

TABLE II. Examination of preparation for delivery

Examine	Defect
Marking	Omitted; incorrect; illegible; not as specified.
Packing	Not as specified.
Contents	Net contents less than specified.
Workmanship	Tears, splits, cracks, holes; closure incomplete or inadequate; leakage or seepage from any part of the container.

4.2.2 Examination of material and workmanship.

4.2.2.1 Lot. For this examination, a lot shall be quebracho extract produced by one manufacturer, at one plant, from the same raw materials, by the same manufacturing process, and offered for examination at one time. Lot size shall be expressed in number of pounds of quebracho extract.

4.2.2.2 Sampling plan. Sample size for this examination shall be one pound. Sample shall be selected by drawing equal portions at random throughout the lot in accordance with table III and combining the portions to form one pound. Defects shall be determined in accordance with table IV. Lot shall be rejected if sample contains one defect.

TABLE III. Sampling for examination of material and workmanship

Lot size (pounds)	No. sample portions	Size sample portions (ounces)
Up to - 1,000	1	16
1,001 - 2,500	2	8
2,501 - 5,000	4	4
5,001 - 10,000	8	2
10,000 - up	16	1

TABLE IV. Examination of material and workmanship

Examine	Defect
Material	Form not as specified.
Workmanship	Color not as specified; foreign matter visibly present.

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4.2.3 Testing.

4.2.3.1 Lot. A lot shall be as specified in 4.2.2.1.

4.2.3.2 Sampling plan. Sample size for testing shall be one pound, selected as specified in 4.2.2.2. Sample portions shall be thoroughly mixed to yield a homogeneous composite and stored in clean, dry, glass container with identifying lot number and date. All testing shall be performed on the composite sample. Lot shall be rejected if the composite fails to meet any specified requirement.

4.3 Test methods. If approved by the Government, test methods which differ from those specified may be used if they provide the minimum quality assurance as specified. In case of dispute as to test results, test methods specified herein shall govern.

4.3.1 Test solution. Into a clean, one-liter volumetric flask, quantitatively transfer 5.9 ± 0.1 grams of quebracho extract, accurately weighed to the nearest 0.1 mg. Dissolve in a few hundred ml of distilled water, previously adjusted to 23-5° C, and complete dilution with distilled water, maintained at 23-5°C. Mix thoroughly to insure homogeneity of solution and maintain solution temperature at 23-5°C throughout testing.

4.3.2 Water Percent water shall be determined in accordance with ALCA Method A20, except that the specimen shall consist of 100 ml of the test solution in 4.3.1.

4.3.3 Insolubles. Percent insolubles shall be determined in accordance with ALCA Method A21, except that the specimen shall consist of 100 ml of the test solution in 4.3.1.

4.3.4 Tannin. Percent tannin shall be determined in accordance with ALCA Method A22, except that the specimen shall consist of 100 ml of the test solution in 4.3.1.

5. PREPARATION FOR DELIVERY

5.1 Packing. Unless otherwise specified (see 6.2), not less than 100 pounds of quebracho extract shall be packed in containers to assure carrier acceptance and safe arrival at destination. Containers shall be in compliance with Uniform Freight or National Motor Freight Classification rules, as applicable.

5.2 Marking. In addition to any special marking required in procurement document (see 6.2), marking of shipping containers shall be in accordance with Fed. Std. No. 123 for civil agencies or MIL-STD-129 for military activities, as applicable.

6. NOTES

6.1 Intended use. Cold-water-soluble quebracho extract is used to treat feedwater in boiler systems which run at pressures not higher than about 300 psi and at ratings up to about 200 percent. It collects mineral deposits and increases the fluidity of boiler sludge which can be more easily removed by periodic blow-down. It also serves as a corrosion inhibitor by scavenging dissolved oxygen from the feedwater. Quebracho is most effective when the alkalinity of the boiler water is maintained at pH 11-12 and it is frequently used as a supplementary treatment along with caustic soda (O-S-598) or soda ash (O-S-571) to improve calcium phosphate sludge where necessary. Types I and II are interchangeable for the same intended use. Type I is a more convenient form to handle.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of quebracho extract required (see 3.1).
(c) Whether special packing is required (see 5.1).
(d) Marking document required (see 5.2).
(e) Whether special marking is required (see 5.2).

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6.3 Supersession data. Interim Federal Specification O-E-00950 contained requirements essentially the same as type I of this document. This revision includes requirements for the solid (lump) form, designated as type II.

Preparing Activity:

GSA-FSS

Civil Agency Coordinating Activities:

GSA-FSS
VA-DMS
INT-MIN

DOD COORDINATION HAS BEEN WAIVED.

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Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

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